SOURCE: MSC.252(83).pdf
data from the same source.

5.4.2 Consistent common reference point

5.4.2.1 The INS should use a single consistent co mmon reference point for all spatially related information. For consistency of measured rang es and bearings, the recommended reference location should be the conning position. Alterna tive reference locations may be used where clearly indicated or distinctively obvious. The sel ection of an alternative reference point should not affect the integrity monitoring process.

5.4.3 Consistency of thresholds

5.4.3.1 The INS should support the consistency of thresholds for monitoring and alert functions.

5.4.3.2 The INS should ensure by automatic means that consistent thresholds are used by different parts of an INS, where practicable.

5.4.3.3 A caution may be given when thresholds entered by the bridge team differ from thresholds set in other parts of the INS.

5.5 Integrity monitoring

5.5.1 The integrity of data should be monitored and verified automatically before being used, or displayed.

5.5.2 The integrity of information should be verified by comparison of the data derived

---CHUNK---

SOURCE: MSC.252(83).pdf
5.5 Integrity monitoring

5.5.1 The integrity of data should be monitored and verified automatically before being used, or displayed.

5.5.2 The integrity of information should be verified by comparison of the data derived independently from at least two sensors and/or sources, if available.

5.5.3 The INS should provide manual or automatic me ans to select the most accurate method of integrity monitoring from the available sensors and/or sources.

5.5.4 A clear indication of the sensors and sources of data selected for integrity monitoring should be provided.

5.5.5 The INS should provide a warning, if integr ity verification is not possible or failed.

5.5.6 Data which fails the integrity monitoring f unction or data where integrity monitoring is not possible should not be used for automatic control systems/functions.

5.6 Marking of data

5.6.1 The data should be marked with the source a nd the results of validity, plausibility checks

---CHUNK---

SOURCE: MSC.252(83).pdf
not possible should not be used for automatic control systems/functions.

5.6 Marking of data

5.6.1 The data should be marked with the source a nd the results of validity, plausibility checks and integrity monitoring to enable subsequent functions to decide whether their input data complies with their requirements or not.

5.7 Selection of sensors and sources

5.7.1 INS should provide two user selectable sensor/source selection modes when multiple sensors/sources are available; manual sensor/sou rce selection mode and automatic sensor/source selection mode.

5.7.2 In manual sensor/source selection mode it should be possible to sele ct individual sensors/ sources for use in the INS. In case a more su itable sensor/source is available this should be indicated.

5.7.3 In automatic sensor/source selection mode, the most suita ble sensors/sources available should be automatically selected for use in the INS. It should further be possible to manually exclude individual sensors/sources from being automatically selected.

Module B – Task related requirements for Integrated Navigation Systems

6 Operational requirements

---CHUNK---

SOURCE: MSC.252(83).pdf
exclude individual sensors/sources from being automatically selected.

Module B – Task related requirements for Integrated Navigation Systems

6 Operational requirements

6.1 The design of the INS should ease the workload of the bridge team and pilot in safely and effectively carrying out the navigation functions incorporated therein.

6.2 The integration should provide all functions , depending of the task for which the INS is used and configured, to facilitate the tasks to be performed by the bridge team and pilot in safely navigating the ship.

6.3 Each part of the INS should comply with all applicable requirements adopted by the Organization, including the requirements of these performance standards.

6.4 When functions of equipment connected to th e INS provide facilities in addition to these performance standards, the operation and, as far as is reasonably practica ble, the malfunction of such additional facilities shoul d not degrade the performance of the INS below the requirements of these standards.

6.5 The integration of functions of individual equipment into the INS should not degrade the

---CHUNK---

SOURCE: MSC.252(83).pdf
such additional facilities shoul d not degrade the performance of the INS below the requirements of these standards.

6.5 The integration of functions of individual equipment into the INS should not degrade the performance below the requirements specified for the individual equipment by the Organization.

6.6 Alerts should be generated and presented according to Module C.

7 Task and functional requirements for an INS

7.1 General

7.1.1 The configuration of the INS should be m odular and task - orie nted. The navigational tasks of an INS are classified as “Route plan ning”, “Route monitoring”, “Collision avoidance”, “Navigation control data”, “Statu s and data display” and “Alert management”. Each of these tasks comprises the respective functions and data.

7.1.2 All tasks of an INS should use the same electronic chart data and other navigational databases such as routes, maps, tide information.

---CHUNK---

SOURCE: MSC.252(83).pdf
tasks comprises the respective functions and data.

7.1.2 All tasks of an INS should use the same electronic chart data and other navigational databases such as routes, maps, tide information.

7.1.3 If Electronic Navigational Charts (ENCs) ar e available, they should be used as common data source for INS.

7.1.4 Paragraphs 7.2 to 7.5 and 7.7 apply, if the respective task is integrated into the INS.

7.2 Task “Route planning”

7.2.1 ECDIS performance standards related mandatory functions and data

The INS should provide the route planning functions and data as specified in Module A and B of the revised ECDIS performance standards (resolution MSC.232(82)).

7.2.2 Procedures for voyage planning

The INS should be capable of supporting procedur es for relevant parts of voyage planning, as adopted by the Organization1.

7.2.3 Additional mandatory functions

The INS should provide means for

- administering the route plan (store a nd load, import, export, documentation, protection),
- having the route check agai nst hazards based on the planned minimum under keel clearance as specified by the mariner,

---CHUNK---

SOURCE: MSC.252(83).pdf

- administering the route plan (store a nd load, import, export, documentation, protection),